

Tutorial 7

Subspaces, linear combinations and spans

1. Let $V = \mathbb{R}^3$, $u = (2, -4, 5)$, $v = (-1, 2, 3)$. With its usual operations of addition and scalar multiplication, V is a vector space. For each of the following statements, say whether it is true or false. Give a reason for your answer.
 - (a) $S = \{\lambda u + v : \lambda \in \mathbb{R}\}$ is a subspace of V .
 - (b) $v \in \text{span}(\{u\})$.
2. In an example in lectures, we introduced the vector space $F(\mathbb{R})$ of all real-valued functions defined on $\mathbb{R}$. In the same way, we can show that $F[0, 4]$, the set of all real-valued functions defined on the interval $[0, 4]$, becomes a vector space if we define addition and scalar multiplication by

$$(f + g)(x) = f(x) + g(x) \text{ and } (\lambda f)(x) = \lambda f(x) \text{ for every } x \in [0, 4],$$

where $f, g \in F[0, 4]$ and $\lambda \in \mathbb{R}$. (See Example 2.1.9 in the notes.)

For each of the subsets of $F[0, 4]$ described in (a) to (d) below, say whether

- (i) it is closed under the usual addition in $F[0, 4]$;
- (ii) it is closed under the usual scalar multiplication in $F[0, 4]$;
- (iii) it is a subspace of $F[0, 4]$

Give reasons for your answers.

- (a) The set of all continuous functions in $F[0, 4]$.
- (b) The set of all f in $F[0, 4]$ such that $f(4) = 0$.
- (c) The set of all f in $F[0, 4]$ such that $f(0) = 4$.
- (d) The set of all continuous functions f in $F[0, 4]$ such that

$$\int_0^3 f(t)dt = f(3).$$

3. Let $M_{2 \times 2}(\mathbb{R})$ be the vector space of 2×2 real matrices, with the usual addition and scalar multiplication. Which of the following subsets are subspaces of $M_{2 \times 2}(\mathbb{R})$? Give reasons for your answers.
 - (a) $\mathcal{S}_1 = \{A \in M_{2 \times 2}(\mathbb{R}) : A \text{ is an upper triangular matrix}\}$.
[Hint: Recall that a square matrix $A = (a_{ij})$ is upper triangular if $a_{ij} = 0$ for all $i > j$.]
 - (b) $\mathcal{S}_2 = \{A \in M_{2 \times 2}(\mathbb{R}) : A^2 = \mathbf{0}\}$ where $\mathbf{0}$ denotes the 2×2 zero matrix.

4. Let $C = \left\{ \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}, \begin{pmatrix} -5 \\ -6 \\ -4 \end{pmatrix}, \begin{pmatrix} 0 \\ -4 \\ 4 \end{pmatrix} \right\}.$

(a) Is $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ in the span of C in $\mathbb{R}^3$? And $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$?

(b) When will an arbitrary vector $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$ be in the span of C ?

[**Hint:** Use the same approach you used in (a). You should find that $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ is in the span of C if and only if x, y and z satisfy a certain equation. Find this equation. What kind of subset of $\mathbb{R}^3$ does this equation describe? The answer to this question should enable you to picture $\text{span}(C)$, and help with the following questions.]

(c) If possible, write one of the vectors in C as a linear combination of the other vectors in C ; if not, say why it cannot be done.

(d) Find, if possible, a vector $\mathbf{v} \in \mathbb{R}^3$ such that $\text{span}(C \cup \{\mathbf{v}\}) = \mathbb{R}^3$. If it is not possible, explain why not.

(e) Can you find a vector $\mathbf{u} \in \text{span}(C)$ such that $\text{span}(C \cup \{\mathbf{u}\}) = \mathbb{R}^3$? Give a reason for your answer.

5. Let $B = \{2x^3, 4x^3 + 3x^2, 3x^3 + 2x^2 + 2, x^3 + 4x^2 + 3x + 5\}$. Is $\text{span } B = P_3$? Give a reason for your answer.

6. Let $S = \left\{ \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix} \right\}.$

(a) Find $\text{span } S$. Do not give the definition only; find an equation describing the matrices in $\text{span } S$. Your answer should have the form

$$\text{span } S = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) : \dots\dots\dots \right\},$$

with an equation containing a, b, c and d in the place of the dots.

(b) Give, if possible, a matrix in $M_{2 \times 2}(\mathbb{R})$ which is not in $\text{span } S$.

7. Let V be a vector space and S and T two subspaces of V . Which of the following subsets of V are subspaces of V ? Give reasons for your answers.

(a) $S \cap T = \{\mathbf{x} \in V : \mathbf{x} \in S \text{ and } \mathbf{x} \in T\}.$

(b) $S \cup T = \{\mathbf{x} \in V : \mathbf{x} \in S \text{ or } \mathbf{x} \in T\}.$

Finally, for those of you who want a bit of a challenge, we have the following:

8. Let S and T be subspaces of a vector space V . Prove or disprove:

- (a) $S + T = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in S, \mathbf{v} \in T\}$ is a subspace of V ,
- (b) $S + T$ is the smallest subspace of V containing $S \cup T$.